Individual Student Contribution Summary — Student 1

Student Name: Atharva Jadhav
Institution: MIT Academy of Engineering (MITAOE)
Project Title: NetSage AI: Automated Cisco Packet Tracer Diagnostic & Troubleshooting Framework
Assigned Project Role: Network Topology Engineer & Packet Tracer Fault Injection Lead

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1. Executive Summary of Contributions

As **Student 1 (Network Topology Engineer)**, my primary responsibility was building the physical network simulation foundation in Cisco Packet Tracer, verifying baseline end-to-end connectivity, injecting live real-world network faults for 12 core cases (`CASE-001` through `CASE-012`), and capturing 36 live evidence screenshots.

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2. Detailed Task Breakdown & Deliverables Owned

Task 1: Enterprise Campus Network Topology Design (`.pkt` File)

- Designed and built the complete Packet Tracer topology file.
- Configured a dual-subnet enterprise campus network topology containing:
  - `Router0` **(Cisco 2811 / 4331):** Configured interface `Gig0/0/0` (`192.168.1.1/24`) and `Gig0/0/1` (`192.168.2.1/24`).
  - `Switch0` **&** `Switch1` **(Cisco 2960 Switches):** Configured access ports `Fa0/1` and 802.1Q inter-switch trunk link `Fa0/23` .
  - `PC0` **&** `PC1` **(End Hosts):** Configured static IP settings on `PC0` (`192.168.1.10/24` , Gateway `192.168.1.1`) and `PC1` (`192.168.2.10/24` , Gateway `192.168.2.1`).

Task 2: Baseline Connectivity Verification (`CASE-000`)

- Verified the initial operational working state (`CASE-000`) before injecting any faults.

- Verified that `PC0` successfully pings `PC1` across `Router0` with 0% packet loss (4/4 ping replies received).
- Documented baseline configuration tables and command outputs.

Task 3: Live Fault Injection & Execution (`CASE-001` to `CASE-012`)

- Injected 12 distinct network fault scenarios live into Packet Tracer across OSI Layers 1–7:
  - **Layer 1 (Physical & Status):** Router interface `Gig0/0/0` shutdown (`CASE-001`), Switch access port `Fa0/1` shutdown (`CASE-005`), Uplink port `Fa0/24` shutdown (`CASE-007`), Trunk port `Fa0/23` shutdown (`CASE-012`).
  - **Layer 2 (Switching & Trunking):** Missing VLAN 20 on inter-switch trunk allowed list (`CASE-006`).
  - **Layer 3 (IP & Gateway):** Default gateway mismatch on `PC0` (`CASE-002`), Subnet mask mismatch on `PC1` (`CASE-003`), Duplicate IP conflict (`CASE-004`), Wrong host IP subnet on `PC1` (`CASE-008`), Missing default gateway (`CASE-009`).
  - **Layer 4 (ACL Security):** Outbound extended ACL 100 blocking ICMP ping traffic (`CASE-010`).
  - **Layer 7 (Services):** Unassigned default gateway setting in Router DHCP pool (`CASE-011`).

Task 4: Live Screenshot Evidence Collection (`evidence/`)

- Collected **36 high-resolution live screenshots** across `evidence/CASE-001/` through `evidence/CASE-012/` and `evidence/baseline/` .
- Captured 3 distinct screenshots per live case:
  1. `01_ping_failure.png` — Showing initial communication breakdown.
  2. `02_evidence_command.png` — Showing diagnostic CLI command output (`show ip interface brief` , `ipconfig` , `show access-lists` , etc.).
  3. `03_ping_success_fix.png` — Showing successful ping verification after applying remediation steps.

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3. Primary Owned File Deliverables

1. `packet_tracer/NETSAGEPACKETTRACER(BASE).pkt` — Live Packet Tracer topology file.
2. `evidence/` **Directory** — 36 live screenshots across 12 live cases + baseline state.

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4. Verification & Testing

- Tested and verified all 12 live cases inside Cisco Packet Tracer.
- Applied exact CLI fix steps for every injected fault and confirmed 100% restoration of ping connectivity (`4/4 ping replies received`).

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5. Summary of Impact

My contributions provided the **factual physical foundation and empirical proof** for the entire project. Without the live topology, fault injection, and screenshot evidence, the AI prompt testing and rule checker verification would have lacked real-world networking ground truth.